

Simulation and Calculations Report

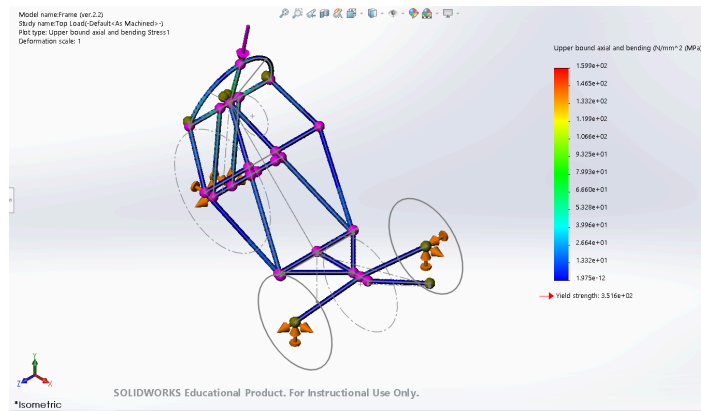


Figure 1: Isometric view of stress analysis resulted from top load simulation

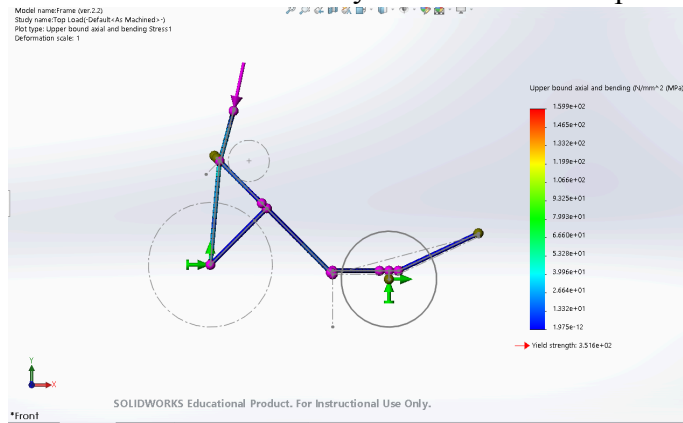


Figure 2: Side view of stress analysis resulted from top load simulation

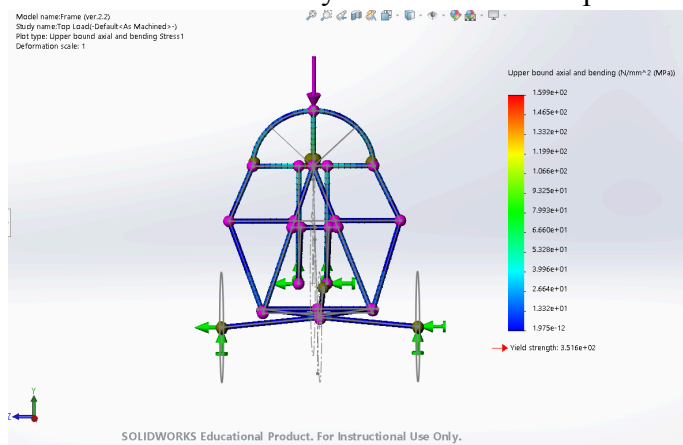


Figure 3: Front view of stress analysis resulted from top load simulation

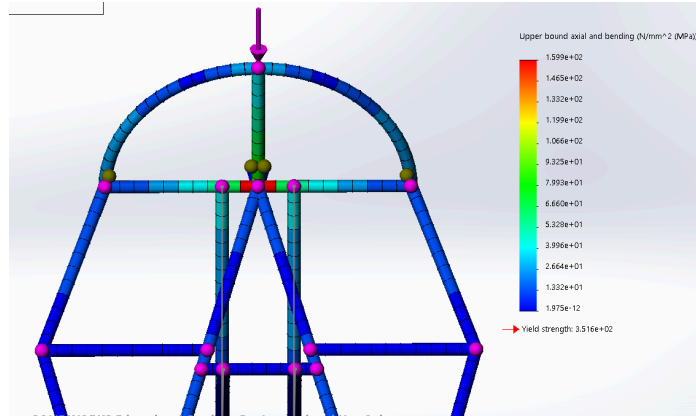


Figure 4: Closed-up front view of stress analysis resulted from top load simulation

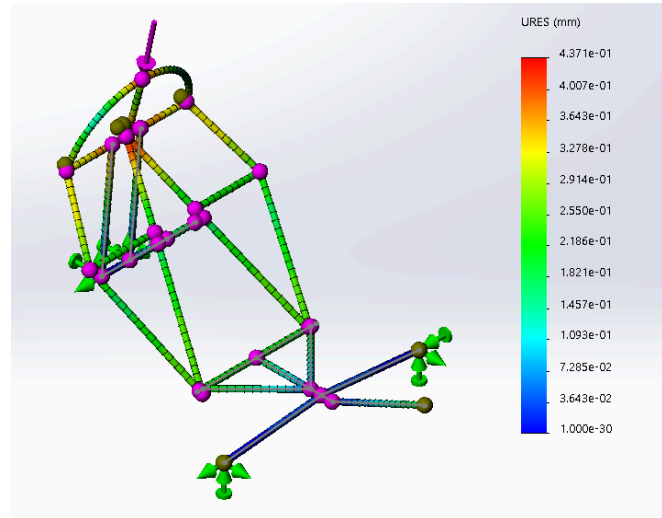


Figure 5: Isometric view of deflection analysis resulted from top load simulation

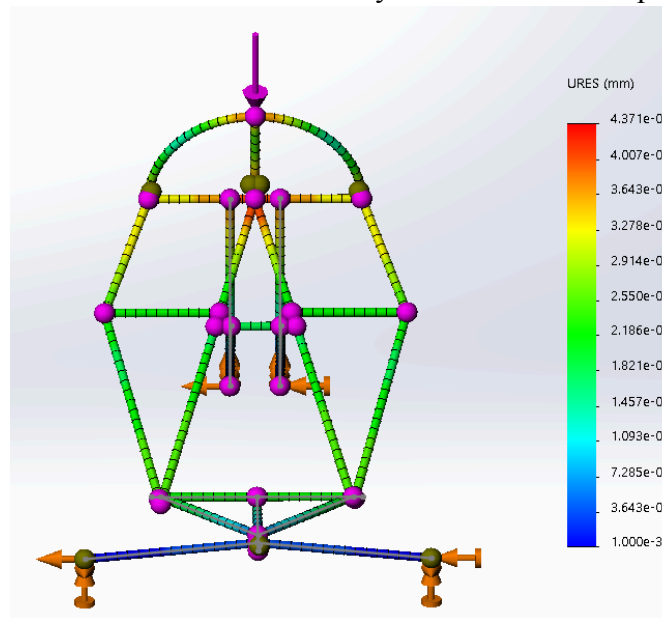


Figure 6: Front view of deflection analysis resulted from top load simulation

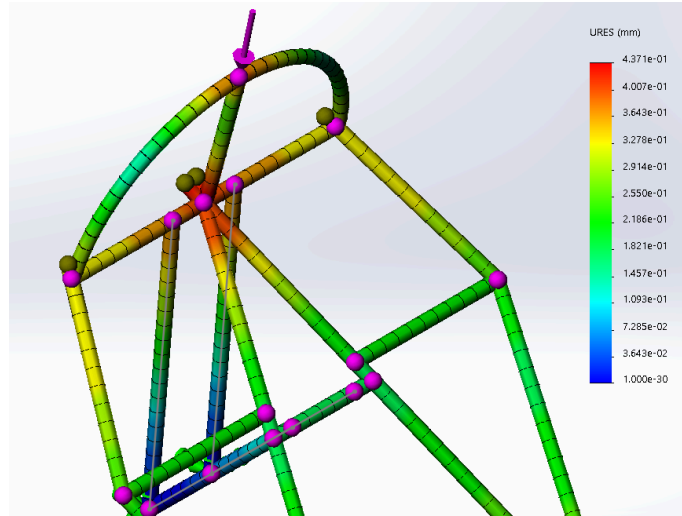


Figure 7: Closed-up isometric view of deflection analysis resulted from top load simulation

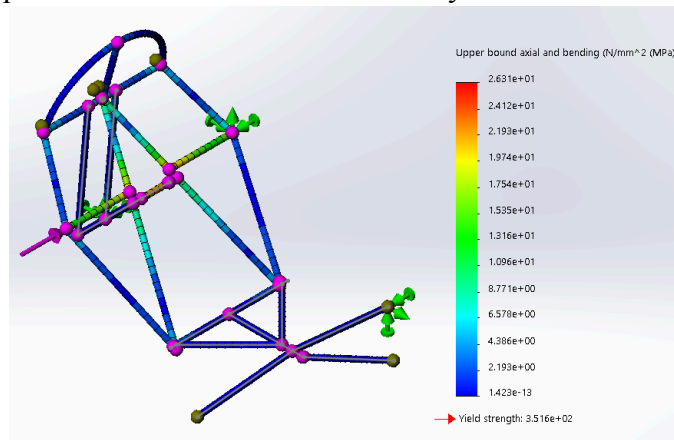


Figure 8: Isometric view of stress analysis resulted from side load simulation (most likely scenario)

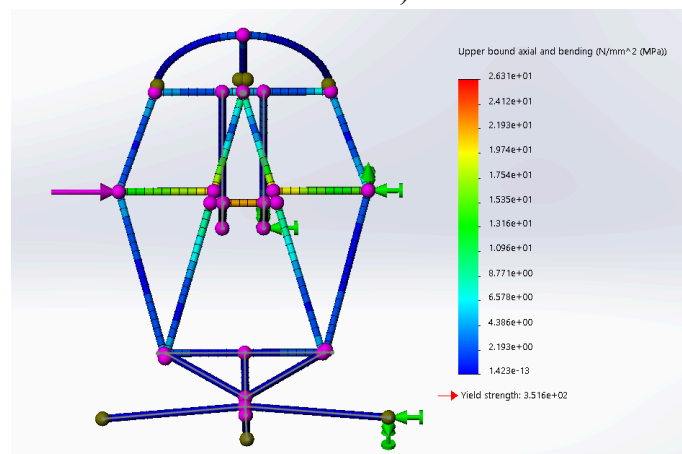


Figure 9: Front view of stress analysis resulted from side load simulation (most likely scenario)

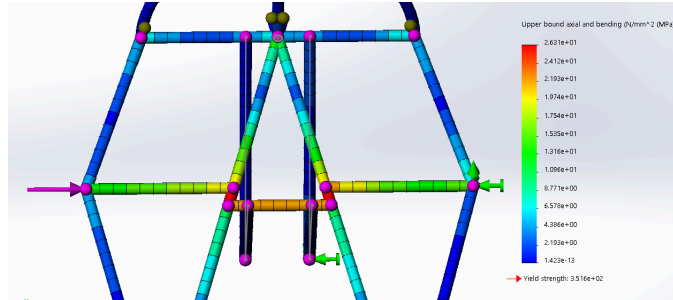


Figure 10: Closed-up front view of stress analysis resulted from side load simulation (most likely scenario)

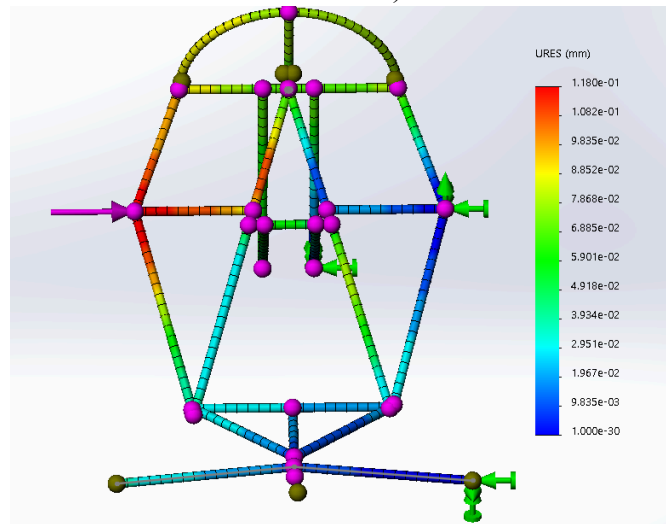


Figure 11: Front view of deflection analysis resulted from side load simulation (most likely scenario)

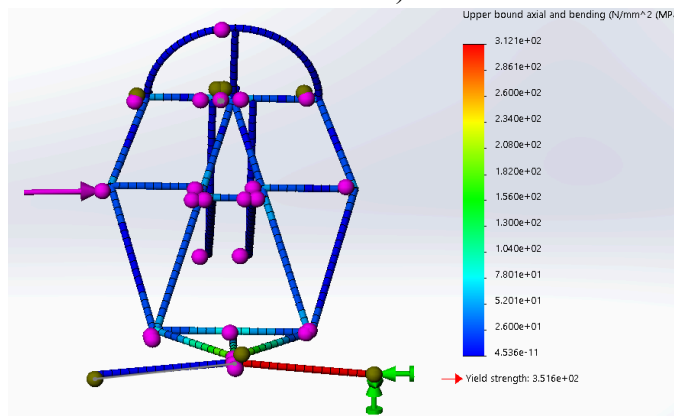


Figure 12: Front view of stress analysis resulted from side load simulation (worst scenario)

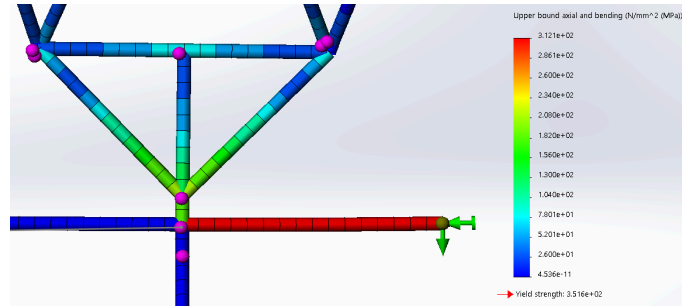


Figure 13: Closed-up top view of stress analysis resulted from side load simulation (worst scenario)

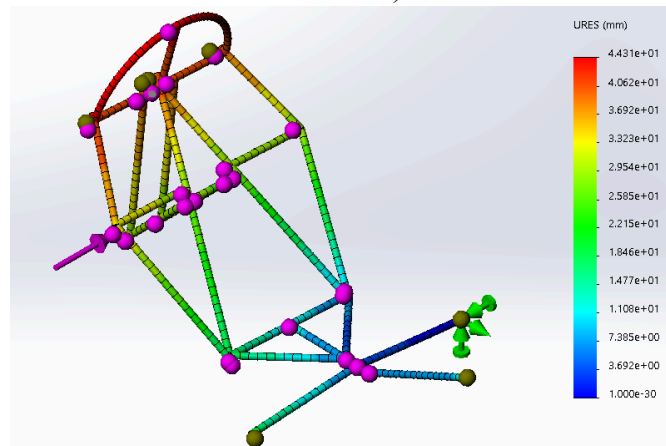


Figure 14: Isometric view of deflection analysis resulted from side load simulation (worst scenario)

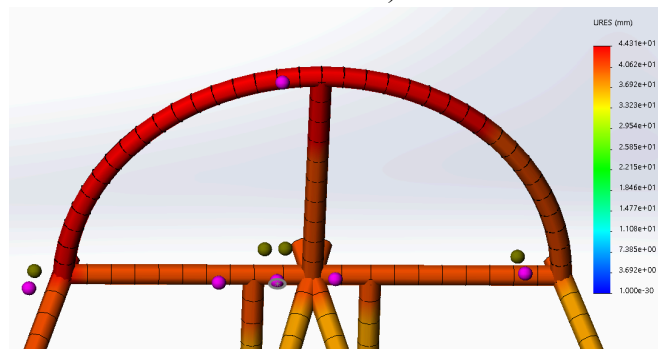


Figure 15: Closed-up front view of deflection analysis resulted from side load simulation (worst scenario)

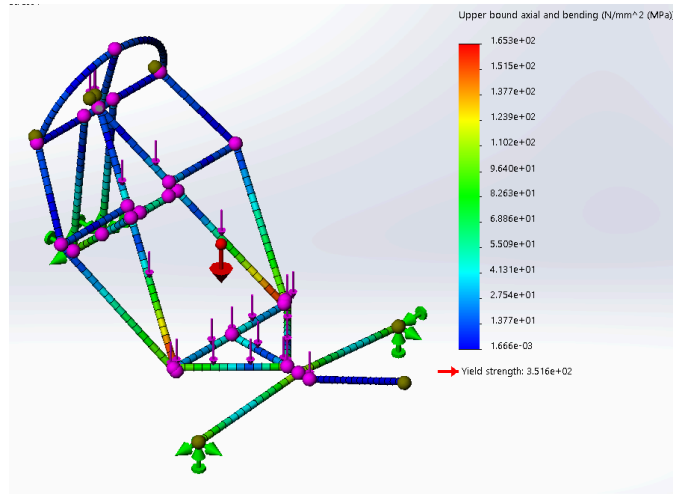


Figure 16: Isometric view of stress analysis resulted from mass load simulation (back and bottom beams)

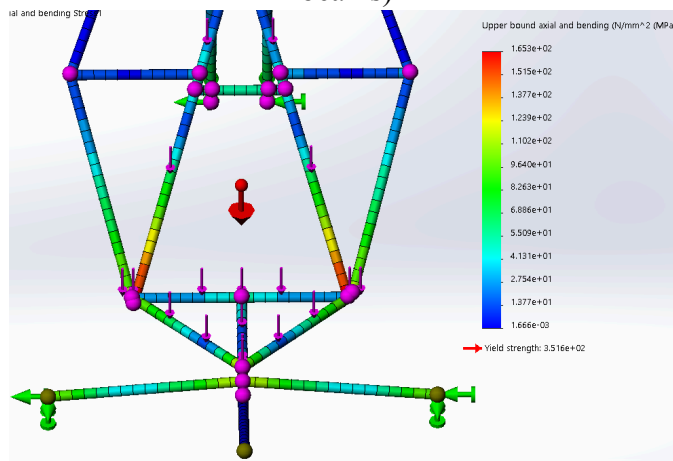


Figure 16: Closed-up front view of stress analysis resulted from mass load simulation on back and bottom beams

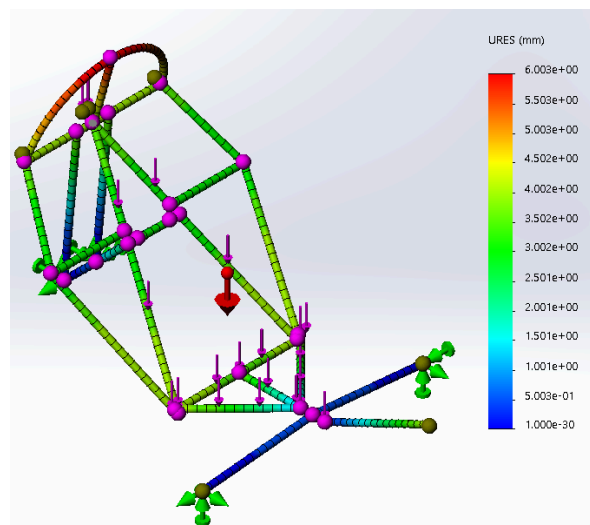


Figure 17: Isometric view of deflection analysis resulted from mass load simulation on back and bottom beams

Table 1: Steel A513 (AISI 1020) Properties

Material	E (GPa)	σ_{yield} (MPa)	σ_{tensile} (MPa)
Steel A513 (AISI 1020)	200	351.571	420.507

Table 2: Simulation Results

Load Type	Load (N)	σ_{max} (MPa)	y_{max} (mm)	FoS	Deflection Requirement
Top	2670	159.9	0.4362	2.20	satisfied
Side (three fixture locations)	1330	26.31	0.118	13.36	satisfied
Side (one fixture location)	1330	312.1	44.31	1.13	satisfied
Mass (back and bottom beams)	1000	165.3	6.003	2.13	satisfied
Mass (back beams)	1000	99.96	3.542	3.52	satisfied